KY–NASH GAME THEORY // THE CALCULUS OF ADMISSIBLE TRANSITIONS

From Preference Maximization to Pre-Execution Consequence Gating, NP → $\mathcal{O}(1)$ Collapse & 205-Byte Witness Equilibria

$G_K = (N, X, \Phi, \Pi_K, \Omega, W, U)$ | $a_{t+1} = \Omega(\Pi_K(\Phi(s_t, i_t))) \in \{\text{OPEN}, \text{HOLD}, \text{KILL}\}$ | $\det(I - \gamma J_{\text{Nash}}) = 1$

Series: Granitum Physics // Sovereign Game Theory (v2.0) **Author:** Marius E. Torjusen (0xABCDEF00, Norway) **Date:** 7 November 2026

1. Ontological Foundation: A Game Begins Not With Players, But With a Gate

Classical game theory assumes that an intended choice $a \in A_i$ is automatically a realized move. **KY–Nash Game Theory** asks a prior, harder question: *which candidate intentions are structurally admissible to become realized consequence?* A player $P_i = (A_i, K_i, \Omega_i, W_i)$ is a **gate-bearing agent** bounded by structural capacity K_i, consequence gate Ω_i, and append-only witness history W_i. Between raw intention i_t and realized history H_T stands a four-stage realization grammar:

$$a'_{t+1} = \Phi(s_t, i_t) \xrightarrow{\text{Structure Test}} \tilde{a}_{t+1} = \Pi_K(a'_{t+1}) \xrightarrow{\text{Consequence Gate}} \Omega(\tilde{a}_{t+1}) \in \{\textbf{OPEN}, \textbf{HOLD}, \textbf{KILL}\}$$

An inadmissible action ($\circ \to \bullet = \textbf{KILL}$) is not a "bad move"—it is discarded boundary residual. Only transitions satisfying $\Omega(\Pi_K(\Phi(s_t, i_t))) = \textbf{OPEN}$ enter the realized game history H_T. Anticipating Kurt Gödel's March 1956 letter to von Neumann by a year, **John Forbes Nash Jr. (January 1955 NSA letter, declassified 2012)** formulated the exponential-search vs. polynomial-verification split: *"The nature of this conjecture is such that I cannot prove it... Nor do I expect it to be proven. But this does not destroy its significance."* KY–Nash operationalizes Nash's 1955 asymmetry: Φ bears the **NP** search ($\exists w$); Ω verifies in $\mathcal{O}(1)$.

2. The Four Strategic Zones (u_i vs. Π_K, Ω)

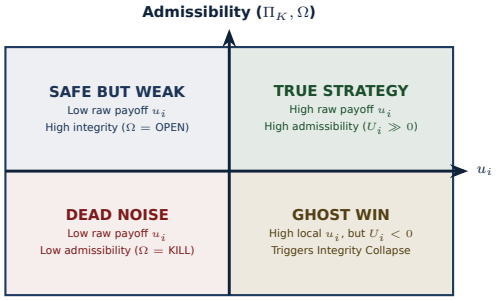


Figure 1: A *Ghost Win* forces a high-u_i transition that destroys the medium of future play ($U_i = u_i - C_K - S_{\text{irr}} - R_W < 0$).

3. Integrity-Adjusted Payoff & NP → $\mathcal{O}(1)$ Collapse

Payoff without a surviving medium is waste heat. KY–Nash replaces raw local utility $u_i(s)$ with the **Integrity-Adjusted Functional** coupled to the Nash resolvent:

$$U_i(s) = u_i(s) - C_K(s) - S_{\text{irr}}(s) - R_W(s), \quad \det(I - \gamma J_{\text{Nash}})^{-1}$$

Quelling Gödel's 1956 NP Dynamics: An **NP** language $x \in L \iff \exists w : V(x, w) = 1$ explodes if the gate searches for w. KY–Nash strips $\exists w$ from the gate: external actors (Φ) bear all **NP**/**PPAD** search costs and must submit a fixed **205-byte witness capsule** (119 B W_{pre} + 86 B W_{post}). The hardware gate (Ω) executes only 1-cycle $\mathcal{O}(1)$ verification $V(a'_{t+1}, w)$ inside an Alexandrov budget $\Delta t_{\text{budget}} = \frac{2r_{\max}}{c} = 133.43$ ns ($r_{\max} = 20$ m) and cognitive airlock $N_A = \lceil 442.892 \text{ MHz} \times 14.0 \text{ ms} \rceil = 6{,}200{,}488$ cycles, isolating self-referential paradoxes via Belnap's $\mathcal{Q}_{\text{LP}}$ ($\pi_R(\Omega) = \emptyset \implies 0.00$ V DC).

4. Structural Comparison: Classical Game Theory vs. Quantum EPR Games vs. KY–Nash

Dimension	Classical Nash Game	Quantum / EPR Game	KY–Nash Admissible Game
Move Ontology	Unconstrained selection $a_i \in A_i$	Unitary / projective operator $\hat{P}_i(\theta_i)$	Gated transition: $a_{t+1} = \Omega(\Pi_K(\Phi(s_t, i_t)))$
Gate Tri-State	Binary (Chosen / Not Chosen)	Superposition → Projection	{**OPEN**, **HOLD**, **KILL**} (KILL ≻ HOLD ≻ OPEN)
Role of HOLD	Absent (static tree)	Uncollapsed entanglement	**Candidate under continuous cost** ($\Delta t \le \tau_{\max}$)
Equilibrium Condition	$\forall a'_i : u_i(a^*_i, a^*_{-i}) \ge u_i(a'_i, a^*_{-i})$	Quantum Nash / ESS under γ	**No improving OPEN-transition** ($\det(I - \gamma J_{\text{Nash}}) = 1$)
Complexity & Ledger	**PPAD** search; memoryless G_t	Hilbert-space unitary evolution	$\Phi \in$ **NP** external; $\Omega \in \mathcal{O}(1)$ over 205B W_{t+1}

5. L1/L1.5 Hardware Realization (SPEC-416) & Cryptographic Construction of Ω_{ind}

In `fpga_capsule_frame_205b_decoder.sv`, every candidate transition requires an allocation-free **205-byte** frame (`208 B` with RS(208, 205); `247 B` L2 wire frame): $\mathbf{W}_{\text{pre}} = \mathbf{119}$ B (`<QQ32sfBH32s32s`: `seq_id`[8B] + `timestamp_ns`[8B] + `candidate_hash`[32B] + `kernel_drift`[4B] + `gate_decision`[1B] + `reason_code`[2B] + `prev_block_hash`[32B] + `w_pre_hash`[32B]) and $\mathbf{W}_{\text{post}} = \mathbf{86}$ B (`<QQ32sBfB32s`: `seq_id`[8B] + `timestamp_ns`[8B] + `w_pre_hash`[32B] + `hpis_latch`[1B] + `thermo_exhaust_nJ`[4B] + `final_status`[1B] + `w_post_hash`[32B]). In the **Whistleblower's Dilemma** ($\mathcal{F} \nvdash G_{\mathcal{F}}$), an incumbent board P_M monopolizing Ω_{int} traps internal reports in indefinite `HOLD`. Rather than relying on a corruptible human third party, Ω_{ind} is enforced by the 1-cycle involution lock (`pre.w_pre_hash == post.w_pre_hash`) and bounded decay $\Delta t = t_{\text{post}} - t_{\text{pre}} \le \tau_{\max}$ (`hold_max_decay_sec = 30.0`). If P_M stalls beyond $\tau_{\max}$ or alters the record ($a'_{M,3}$, local gain $u_M = +1.00$), the decoder triggers `0xFF TRAP_TAMPER`, enforcing $U_M(a'_{M,3}) = 1.00 - 0.55\,(C_K) - 0.45\,(S_{\text{irr}}) - 0.80\,(R_W) = -0.80 < 0$ (**Ghost Win**).

[THE KY–NASH FUNDAMENTAL CUT // SOVEREIGN AXIOM]

Law I: A move is valid only if it carries its own transition across structure (K), boundary (Π_K), cost ($C_K + S_{\text{irr}}$), and 205B witness (W). **Law II:** **HOLD** is a candidate suspended under continuous cost. **Law III:** *"Let the external world (Φ) bear the* **NP** *search; the gate (Ω) verifies in $\mathcal{O}(1)$ at 0.00 V DC."*

Reference Index: https://github.com/sololys/Infranett | **Forum:** *Granitum Physics — Topology, Thermodynamics & Consequence Engineering*

Sololys (0xABCDEF00)